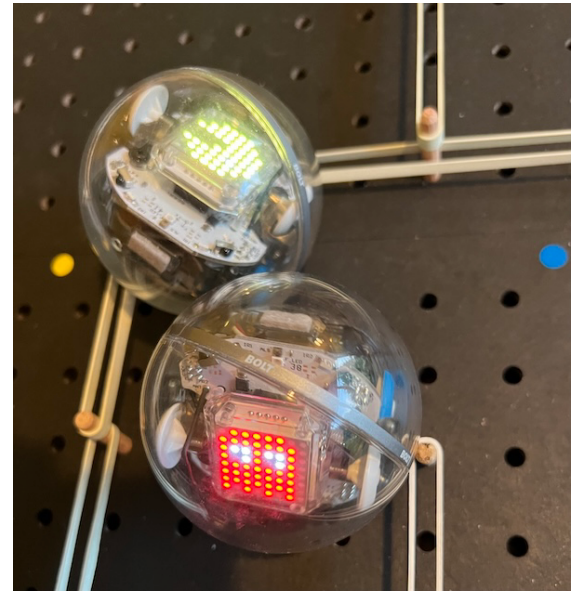




Robot Ghost

You've (probably) played Pac-Man the video game, but have you played it with robots?



This is the original Namco Pac-Man videogame from 1980. Can we recreate it with robots?

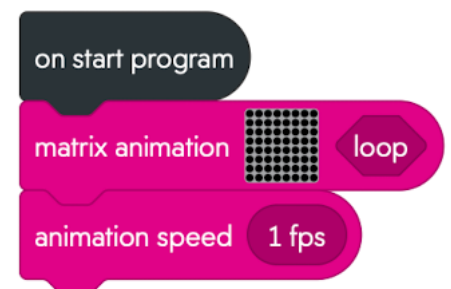
https://youtu.be/i_OjztdQ8iw

CHARACTER	/ NICKNAME	
	- SHADOW	"BLINKY"
	- SPEEDY	"PINKY"
	- BASHFUL	"INKY"
	- POKEY	"CLYDE"

The videogame has four ghostly characters named "Blinky", "Pinky", "Inky" and "Clyde." We need at least one of each character.

- 1) In Chrome, open <https://edu.sphero.com/code>
- 2) Connect your **Sphero BOLT** robot: click 
- 3) Select  then 
- 4) Your BOLT has a sticker with a code on it. Choose the matching BOLT from the list then 
- 5) Check your robot is showing the right image then 
- 6) Click **+ CREATE PROGRAM.** 
- 7) Name your program after the character you're creating. Leave **BLOCKS** selected. Select **SPHERO BOLT**, deselect **SPHERO BOLT+** then **CREATE.**
- 8) Add code for a **matrix animation** with *your ghost* design.

Click on the matrix grid to edit then  to add a new 8x8 matrix. Click  to finish.



9) Animate it to make its eyes look left  then right .

Select your graphic then click  to edit it.

Click **+** to add a new frame.

To remove a frame, click , select a frame then .

10)  your code. Does it work?

When ghosts run away from Pac-Man, they turn blue and take fright.

Add another matrix design for this.

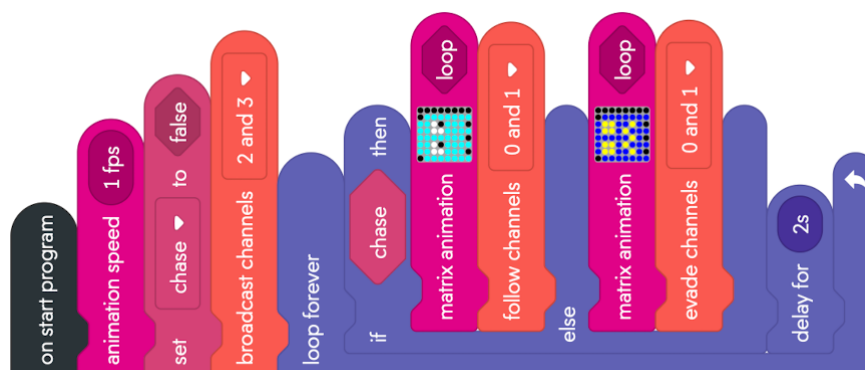


11) In **Variables**, create a new variable called **chase**.

This is **boolean** because it will only be *true* or *false*.

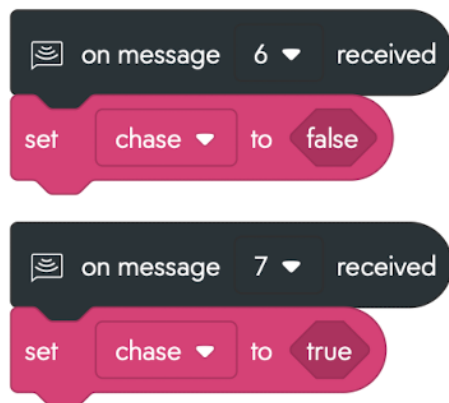
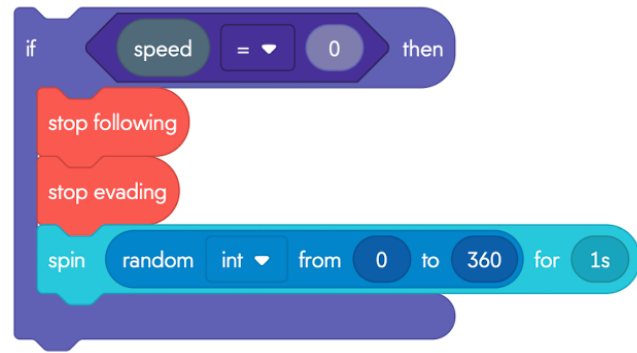
Sphero robots communicate using infra-red light invisible to us. Ghosts broadcast their position on channels 2, 3 (short & long range), and follow Pac-Man on channels 0, 1.

12) Replace the existing code with the following:



Sometimes the robot gets stuck on the maze walls.

13) Add the code (right) at the end of the loop (but still inside the loop) to detect when it has stopped moving and give it a random spin.



14) When Pac-Man receives an **energizer** it sends a message to reverse the **chase**. Add code (left) to receive this message and set chase to **true** or **false**.

*To **Save** your code go back to the main page. Don't forget to add "Blinky", "Pinky", "Inky" or "Clyde" to the name of your program. Click on the 3 dots (ellipsis) next to your program and download to your computer.*